Adversarial attacks

Weekend 4, Saturday morning

Carlos Cotrini

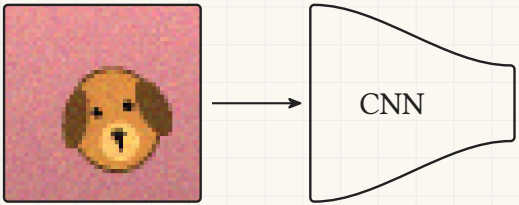
CAS Building ML/AI Applications, ETH Zurich

Saturday 24 October 2026

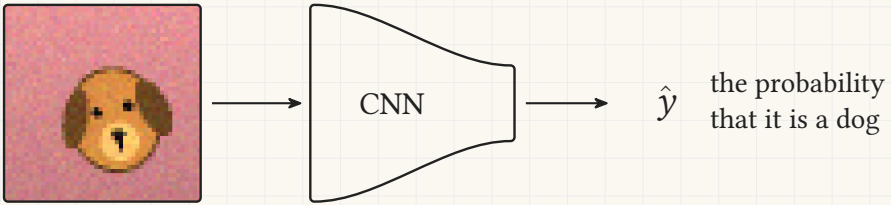
Dog or cat?



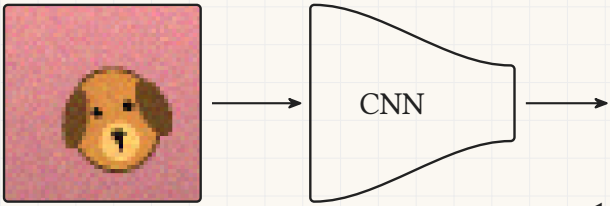
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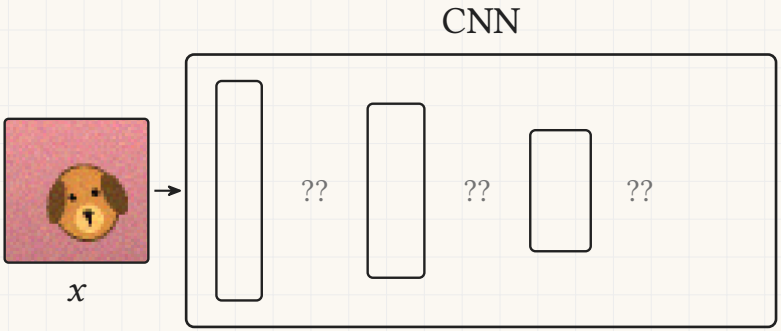
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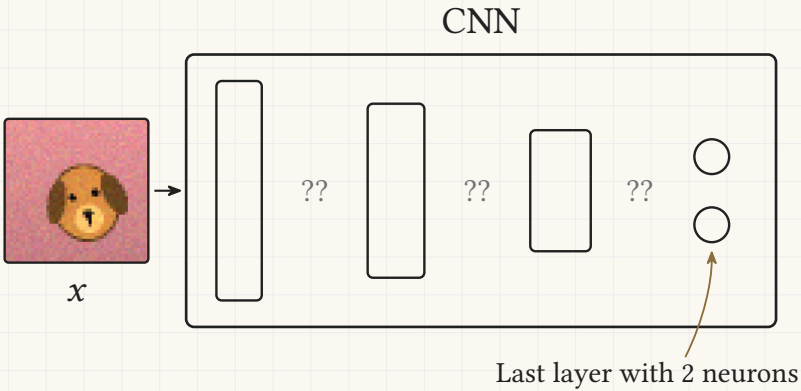
Dog or cat?


CNN
$\hat{y}$
the probability that it is a dog
$1 - \hat{y}$
the probability that it is a cat

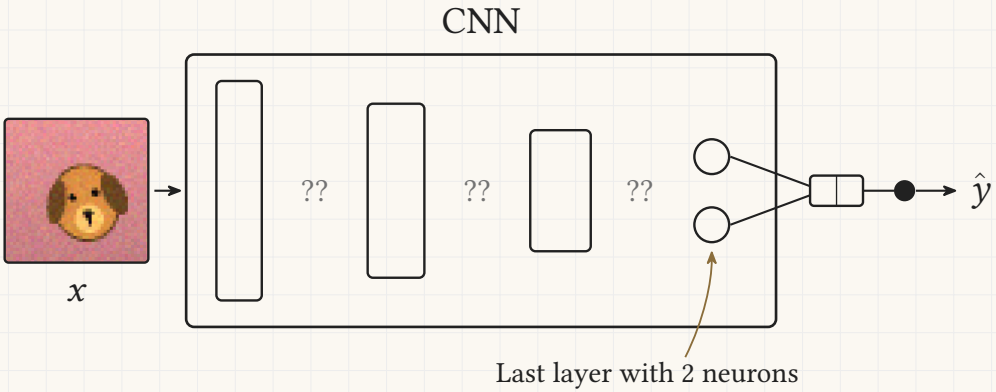
Inside the CNN



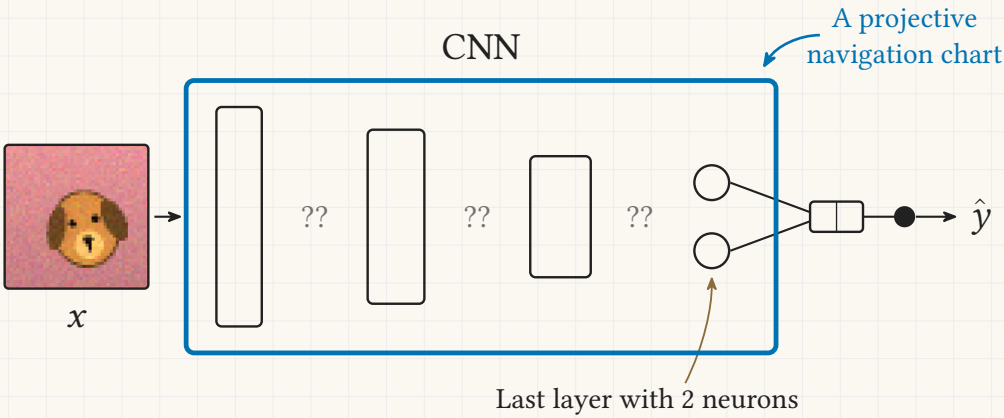
Inside the CNN



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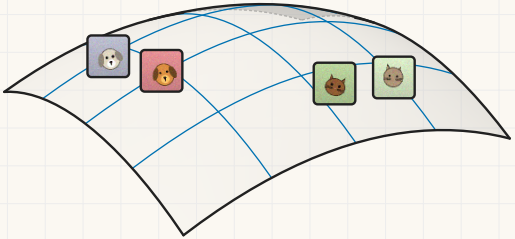


Inside the CNN



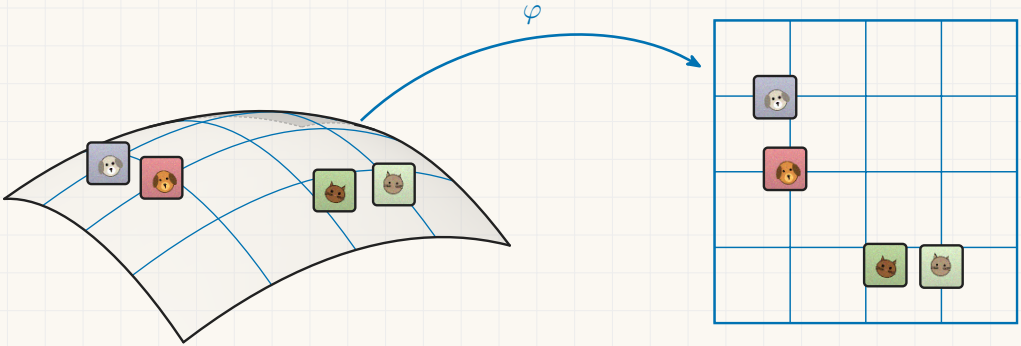
Dogs and cats on a chart

Photos of pets



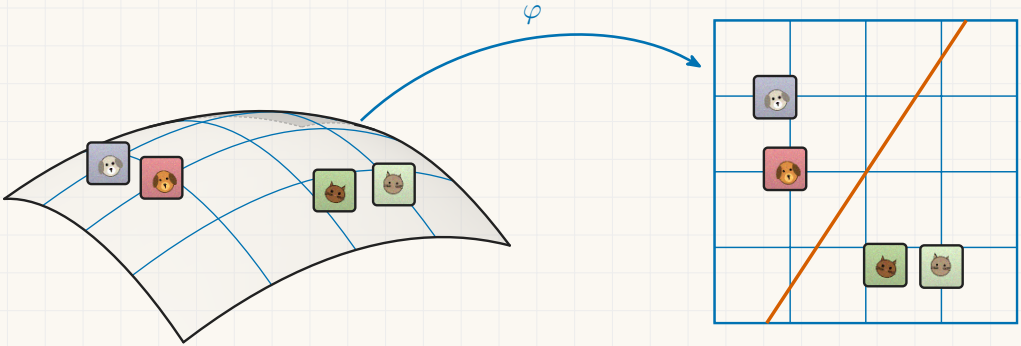
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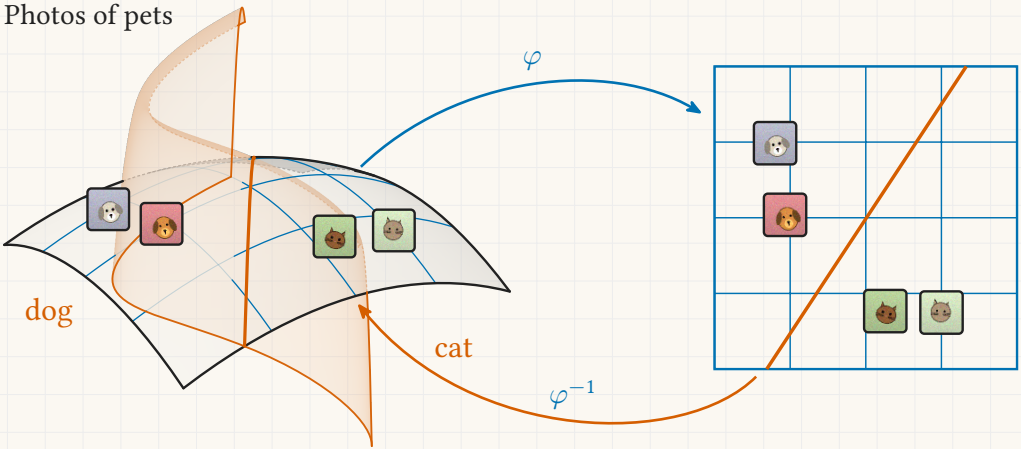


Dogs and cats on a chart

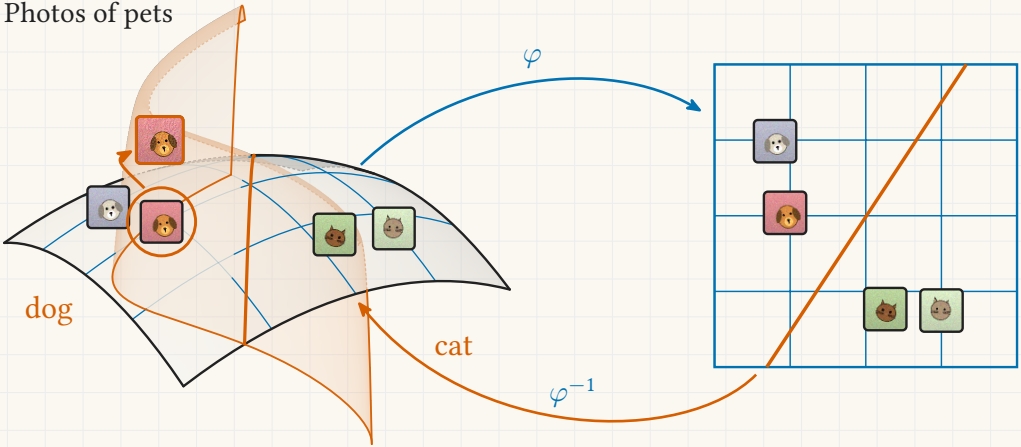
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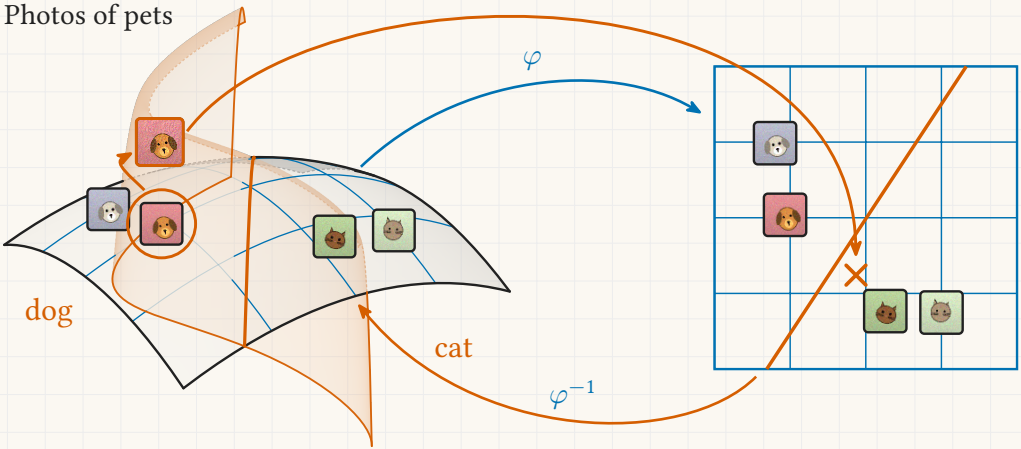
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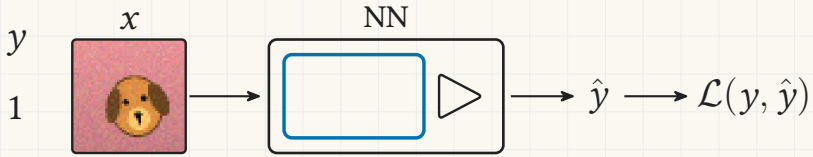
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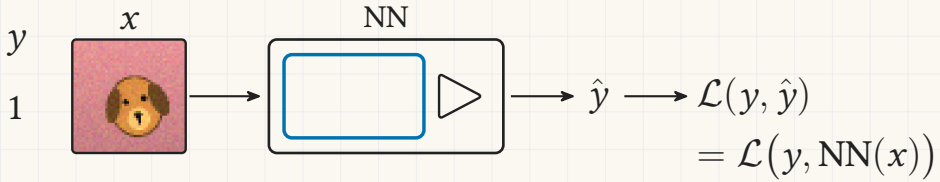
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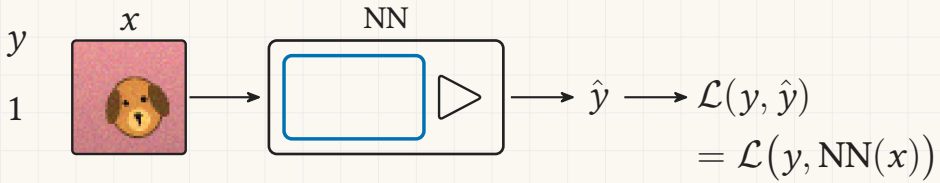
The attack



The attack

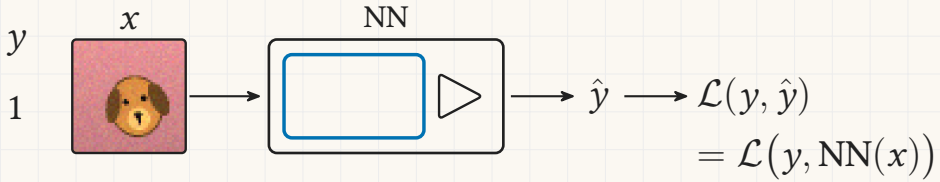


The attack



Goal: change x to change $\hat{y}$.

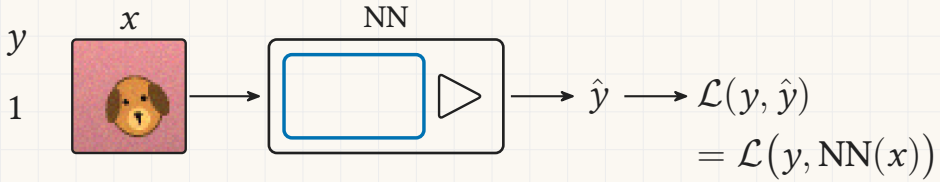
The attack



Goal: change x to change $\hat{y}$.

Add ε to x to increase $\mathcal{L}\big(y, \mathrm{NN}(x)\big)$.

The attack

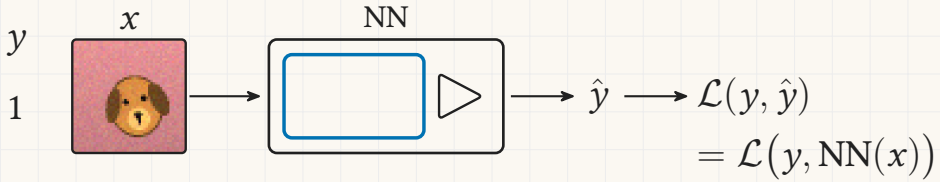


Goal: change x to change $\hat{y}$.

Add ε to x to increase $\mathcal{L}\big(y, \mathrm{NN}(x)\big)$.

$$x = x + \varepsilon \, \nabla_x \, \mathcal{L}\big(y, \mathrm{NN}(x)\big)$$

The attack



Goal: change x to change $\hat{y}$.

Add ε to x to increase $\mathcal{L}\big(y, \mathrm{NN}(x)\big)$.

$$x = x + \varepsilon \, \nabla_x \mathcal{L}\big(y, \mathrm{NN}(x)\big)$$

$$\Rightarrow \qquad x = x + \varepsilon \, \mathrm{sgn}\Big(\nabla_x \mathcal{L}\big(y, \mathrm{NN}(x)\big)\Big)$$